



USER GUIDE

# VirtualGlove Input Modes

Joystick mode, merged physical controls, and native Super Glove Ball input.

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VirtualGlove can control an NES game in two different ways. **Joystick mode** turns recognised gestures and hand movement into ordinary NES controller input. **Native Power Glove mode** gives a compatible game the position-and-pose packet it expects from the original glove.

Most games use joystick mode. Super Glove Ball can use either mode, although native mode provides its distinctive continuous Robo-Glove movement.

This guide explains what the modes do, how physical controllers participate, how to choose the right mode, and what to check when controls do not behave as expected. The [Game and Gesture Guide](#) remains the reference for individual programs and gestures.

## Choose the right mode

| You want to...                                      | Use                        | Emulator                     |
|-----------------------------------------------------|----------------------------|------------------------------|
| Play an ordinary registered NES game                | Joystick mode              | FCEUmm or stock Nestopia     |
| Use Programs 1-14 or A-I                            | Joystick mode              | FCEUmm or stock Nestopia     |
| Use gestures and a physical controller together     | Joystick mode              | FCEUmm or stock Nestopia     |
| Play Super Glove Ball with continuous hand position | Native Power Glove mode    | Nestopia (VirtualGlove)      |
| Play Super Glove Ball as a conventional NES game    | Joystick mode              | FCEUmm                       |
| Temporarily use only the physical controller        | Gestures off or Program 14 | The game's selected emulator |

Changing modes does not require a second camera calibration. Both paths begin with the same camera, selected player, saved centre, hand scale, reach, and gesture recognition.

## How joystick mode works

Joystick mode presents familiar RetroPad controls to FCEUmm or stock Nestopia: D-pad, A, B, Start, and Select. The active profile decides which recognised movement or gesture produces each control. Returning the hand to its saved centre releases movement, and tracking loss safely releases every VirtualGlove control.

The console receives authenticated, current controller state. It does not replay a queue of old hand positions. Rapid-fire settings are applied at the console and can be changed from the Dashboard while the game is running.

Controller Router gives its two input sources deliberately different reach. Assigned physical controllers follow their merged player into every RetroArch/Libretro system. VirtualGlove remains limited to supported NES joystick cores. At one of those NES launches, Router clears any glove state observed in the frontend and waits for a fresh neutral hand before accepting new gestures. Camera position axes are not forwarded through this joystick path; native Super Glove Ball receives them through its separate guarded native-state channel.

## Player 1 on each platform

| Platform                         | How joystick-mode Player 1 works                                                                                                                                                                                                  |
|----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| RetroPie - standard installation | VirtualGlove appears as a separate <b>VirtualGlove</b> gamepad. Existing physical controllers remain separate and continue to work.                                                                                               |
| RetroPie - routed cabinet        | Optional <b>Controller Router</b> can combine configured I-PACs, joypads, and VirtualGlove into <b>VirtualGlove Merged Player 1-4</b> . The standard installation remains unchanged until Router is explicitly saved and applied. |
| Recalbox                         | <b>Controller Router</b> creates enabled merged Players 1-4. The released Player 1 selection migrates automatically, while the original controllers continue to operate EmulationStation.                                         |
| Batocera                         | Uses the same <b>Controller Router</b> model as Recalbox and resolves current Linux and RetroArch indexes for Libretro gameplay.                                                                                                  |
| LaunchBox                        | The physical XInput controller remains Player 1. VirtualGlove joins it through RetroArch's loopback Network RetroPad. The real keyboard remains available.                                                                        |

## Standard RetroPie and the VirtualGlove arcade cabinet

A fresh RetroPie installation does not need a merger. The standard installer creates one **VirtualGlove** gamepad, installs its RetroArch mapping, and leaves the console's existing joypads untouched. This is the arrangement that worked on a clean RetroPie installation and remains the supported default.

The VirtualGlove development cabinet needs routing because it has several ways to control the same players. Its accepted Controller Router configuration combines the two I-PAC gamepad interfaces, supported 8BitDo controllers, and VirtualGlove into two canonical outputs:

- **Arcade Merged Player 1:** I-PAC Player 1 + 8BitDo Player 1 + VirtualGlove.
- **Arcade Merged Player 2:** I-PAC Player 2 + 8BitDo Player 2.

Router rediscovers sources after connection changes instead of saving an **eventN** path. A button stays pressed while any source holds it, and removing a source releases only that source's controls. VirtualGlove contributes NES Select but cannot produce the merged device's dedicated hotkey-enable button.

This cabinet component was authored as part of the VirtualGlove work and is now preserved under [retropie/arcade-cabinet-merger/](#). It was previously deployed directly to [/usr/local/sbin/arcade-gamepad-merger](#), which is why its source was missing from the repository even though it belonged to the project.

The checked-in original merger intentionally remains a reference and one-command rollback source. The cabinet migration has now passed reboot, hotkey, I-PAC, 8BitDo, VirtualGlove, and simultaneous-input testing. Its active path is the same configurable Controller Router used by Recalbox and Batocera.

## Controller Router

Those platforms normally discover a physical controller, read its EmulationStation mapping, and generate a RetroArch assignment at game launch. Trying to add a second Player 1 after that process can change button meanings, move a controller to another index, or let one press reach both RetroArch and a platform hotkey handler.

VirtualGlove therefore creates up to four canonical gameplay devices named **VirtualGlove Merged Player 1-4**:

1. Setup inventories only controllers already configured by EmulationStation. The user reviews suggested frontend-order assignments before anything is saved.

2. The versioned `controller-router.json` records stable identities, the last validated EmulationStation mappings, enabled players, and at most one VirtualGlove player. It never stores `/dev/input/eventN`, `/dev/input/jsN`, or a RetroArch index.
3. At boot, Router finds each saved source, rereads its current validated EmulationStation mapping, and creates only the enabled outputs. A later EmulationStation remap keeps the same player assignment and takes effect automatically while idle or at the next game launch.
4. At game launch, the current mapping, Linux event device, and current RetroArch joystick index are resolved again. USB enumeration may change without changing the selected controller.
5. Router clears any VirtualGlove state seen before launch. Physical sources work immediately; VirtualGlove is admitted after one fresh neutral hand observation, so held frontend input cannot block or steer the new game.
6. During Libretro gameplay, Router takes exclusive ownership of each assigned physical event device. This prevents the original device, the platform hotkey service, and the merged device from interpreting the same press. VirtualGlove input is admitted only for supported NES joystick cores.
7. Buttons remain held while any assigned source holds them. For each physical axis, the latest active source owns it until neutral; another still-active source then resumes. Any non-neutral physical source outranks VirtualGlove.
8. At game exit, every merged state is neutralized before exclusive ownership is released. EmulationStation then continues using the original controller.

The platforms apply those assignments at different supported layers. RetroPie resolves the merged indexes in its launch hook. Batocera persists them through its managed `batocera.conf` settings. Recalbox stores them in `/recalbox/share/roms/.retroarch.cfg`, which Recalbox deliberately incorporates when it generates each Libretro launch. The generated `retroarchcustom.cfg.overrides.cfg` is output, not persistent configuration, and must not be edited directly.

Nestopia (VirtualGlove) keeps gesture input on its distinct native-state path. Its physical controller still participates through the all-Libretro Router assignment, preserving menu and exit hotkeys without duplicating glove input.

The physical controller's own hotkey is mapped to a dedicated merged button. VirtualGlove Select can emit only NES Select; it cannot enable RetroArch hotkeys. This is why Select no longer opens the RetroArch menu while still working in a game.

If a physical controller disconnects during play, only its held inputs are released. VirtualGlove can remain active, and the saved controller reconnects when the same device returns. If several indistinguishable controllers are present, the installer requires an explicit Player 1 choice rather than guessing.

One physical source can belong to only one player, but several physical sources may share a player. The one paired VirtualGlove can be unassigned or assigned to exactly one player. Physical-only configurations are valid and need no pairing. Only Player 1 carries a physical hotkey. Player 2-4 hotkeys are ordinary or ignored controls, and VirtualGlove can never emit the hotkey-enable button.

## Local assignment screen

Run `sudo /opt/virtualglove/bin/virtualglove-controller-router setup` on RetroPie. On Recalbox and Batocera, use `sh` with the persistent `scripts/virtualglove-controller-router setup` launcher inside the VirtualGlove installation because their persistent shares are mounted without direct program execution. The dependency-free screen detects the platform, shows stable controller identities and connection state, and lets the operator assign each source or VirtualGlove to Players 1-4. **Test controls** reports live buttons and directions without launching a game. Nothing changes until **Save and verify** is confirmed, and the prior assignment can be restored from the same screen.

The local screen does not create or replace a pairing credential. Pairing authorizes the VirtualGlove Controller and its web Setup page to reach the console; routing decides which merged player receives each configured physical source and which supported NES player may receive VirtualGlove gestures.

FCEUmm recognises known Four Score games by CRC. Four merged outputs do not make an ordinary game four-player. For a compatible altered ROM that FCEUmm does not recognise, the advanced registry setting `"four_score": "force"` selects its User 5 four-player adaptor; leaving the field out keeps automatic detection.

## What the merger does not change

- It does not rewrite the controller's EmulationStation mapping.
- It follows a valid EmulationStation remap automatically without changing the saved player assignment. An incomplete mapping disables only that source until it is configured correctly.
- It applies enabled physical Player 1-4 assignments to Libretro gameplay while keeping VirtualGlove gesture output limited to its supported NES paths.
- It does not depend on the order in which USB devices appeared after boot.
- It does not make the merged device navigate the frontend.
- It does not let VirtualGlove gestures activate the physical hotkey.

## Test joystick mode

Use a registered game such as Super Mario Bros. with either FCEUmm or stock Nestopia:

1. Confirm that the physical controller works in EmulationStation.
2. Launch the game and wait for the title screen.
3. Use the physical controller immediately to test directions, A, B, Start, and Select. You must not need to stop VirtualGlove first.
4. Briefly rest the hand at neutral, then test the corresponding gestures.
5. Hold a physical direction while making a different VirtualGlove movement. The physical direction should remain authoritative until released.
6. On a routed console, press physical Select by itself. It must behave as Select and must not open the RetroArch menu.
7. Test the platform's physical hotkey combinations, including menu and exit.

After a reboot, repeat steps 2-4. A changed event number or joypad index should not require reconfiguration.

## How native Power Glove mode works

Native mode is available for the exact registered Super Glove Ball game through the separately named **Nestopia (VirtualGlove)** core. Stock Nestopia remains untouched, and FCEUmm remains available as the conventional fallback.

The shared Controller-to-console receiver publishes one coherent, current native-state sample. Nestopia (VirtualGlove) reads at most one sample per emulated frame and converts it into the ten-byte Power Glove packet used by the validated game. It adds no replay queue and no second smoothing layer.

In joystick mode, moving far enough right means "hold Right." In native mode, the calibrated position means "place the Robo-Glove at this X/Y coordinate." That difference is the reason native mode can track continuous movement inside the playfield.

| What you do                     | Confirmed native Super Glove Ball behaviour             |
|---------------------------------|---------------------------------------------------------|
| Move horizontally or vertically | Position the Robo-Glove continuously in X/Y             |
| Open the hand                   | Release or throw                                        |
| Close the hand                  | Grab or catch                                           |
| Point with the index finger     | Fire Robo-Bullets                                       |
| Make a fist and push forward    | Power Punch                                             |
| Use the Start gesture           | Start the game                                          |
| Press physical Player 1 Start   | Start the game through the same confirmed native action |

Every action needed to complete the game has been confirmed in live play. Wrist rotation and unobserved native button fields remain neutral because the validated game has not demonstrated a required action for them. Guessing packet values would risk unintended input.

## Safety and freshness

Missing calibration, the wrong profile, lost tracking, invalid state, or a sample older than 250 milliseconds produces neutral native input. A brief missed observation may retain only X/Y for up to 180 milliseconds while actions release. Recovery accepts aligned forward movement immediately and guards one strongly contradictory or unusually distant result.

The emulator core reported by the console is part of the authenticated game session. Native output is enabled only for the combination of the [super\\_glove\\_ball](#) profile and Nestopia (VirtualGlove). The same ROM in FCEUmm, another core, or an unknown core receives joystick-mode output instead.

## Select native Super Glove Ball

The installer adds Nestopia (VirtualGlove) without replacing another emulator. Exact-ROM selection is automatic where the platform supports it, but an existing per-game choice is preserved.

1. Register the exact Super Glove Ball ROM in Setup.
2. Confirm the console installation check reports the optional native core as available.
3. Choose **Nestopia (VirtualGlove)** for that ROM if the platform has preserved a previous emulator choice.
4. Launch the game before starting the Controller so the emulated glove is attached during the game's detection sequence.
5. Start the Controller and use the Start gesture or physical Start.

To return to conventional controls, select FCEUmm for that ROM. This does not change player calibration, gesture tuning, the physical controller mapping, or the ROM.

## Native support by platform

| Platform  | Native-core installation and selection                                                                                  |
|-----------|-------------------------------------------------------------------------------------------------------------------------|
| RetroPie  | Selects and verifies a packaged core from RetroArch's actual 32- or 64-bit ABI, then offers it as a per-ROM emulator.   |
| Recalbox  | Installs a verified architecture build in persistent storage and exposes it through reversible runtime overlays.        |
| Batocera  | Selects and verifies the packaged architecture build, then exposes it through reversible overlays and a per-ROM choice. |
| LaunchBox | Installs a separately named x86-64 DLL and uses the VirtualGlove RetroArch wrapper for the exact registered ROM.        |

If the native artifact is missing, changed, incompatible, or not selected, VirtualGlove safely uses FCEUmm joystick mode. Installers never replace stock Nestopia.

## Confirmed compatibility boundary

Native behaviour is validated against [Super Glove Ball \(USA\)](#) with SHA-256:

[ad60ef1b62cd1b3bc02a9320376067347a8ab2ebbe46e1616693d8379c9d9a7b](#)

The ROM is not included with VirtualGlove. The exact game reads ten bytes per sample, requires the final validation byte, and consumes native X/Y, signed depth, pose, and Start values confirmed by controlled traces and live play. These results are not a claim that every ROM revision or Power Glove-compatible game uses the same packet fields.

Headless same-state tests found the first visible directional response on frame 3 for both this native path and the same ROM in FCEUmm. That isolates emulator response; it does not include camera exposure, recognition, network delivery, or display latency. The investigation and rejected alternatives are preserved in the [Engineering Journey](#).

## When controls do not work

Check the visible symptom first:

- **Physical controller works in the frontend but not in a Recalbox/Batocera game:** rerun the current console installer check. Confirm that the saved controller is connected, its merged player exists, and the persistent all-Libretro assignment passes. Do not edit Recalbox's generated [retroarchcustom.cfg.overrides.cfg](#).
- **Buttons are rearranged:** exit the game and confirm the controller mapping in EmulationStation. Controller Router adopts a valid remap automatically on the next launch; Setup shows **Mapping refreshed** when it differs from the saved last-known mapping.
- **Select opens the RetroArch menu:** the physical controller is reaching both the platform hotkey service and the merged path. Exit the game and restart the VirtualGlove console service or rerun the current installer.
- **VirtualGlove works but the physical controller disappeared:** reconnect the selected controller. If it was replaced with a different model, rerun the installer and select the new Player 1 device.
- **Super Glove Ball moves like a D-pad:** the game is running in FCEUmm. Choose Nestopia (VirtualGlove) for continuous native movement.

- **Super Glove Ball does not detect native input:** confirm the registered ROM, selected core, calibration, and active Controller status. Then exit and relaunch so the native peripheral is attached before detection.
- **Controls remain held after interruption:** stop the Controller and exit the game. Timeout, disconnect, game-exit, and service-shutdown paths all publish a neutral state; use the [Troubleshooting Guide](#) if they do not.

For installed paths, device records, service checks, and command-line options, see the [Configuration Reference](#). For packet and ROM-level evidence, see the [Power Glove Game Input Audit](#).